4. Software

4.2 Types of programming language, translators and integrated development environments (IDEs)

Marking Scheme

Q1)

1 mark per correctly placed tick

statement	interpreter	compiler
takes one statement at a time and executes it	✓	
generates an error report at the end of translation of the whole program		✓
stops the translation process as soon as the first error is encountered	✓	
slow speed of execution of program loops	✓	
translates the entire program in one go		✓

[5]

Q2)

(a) 1 mark per correctly placed tick

statement	interpreter	compiler
creates an executable file that runs directly on the computer		✓
more likely to crash the computer since the machine code produced runs directly on the processor		✓
easier to debug since each line of code is analysed and checked before being executed	✓	
slow speed of execution of program loops	✓	
it is more difficult to modify the code since the executable code is now in machine code format		✓

[5]

(b) Any **one** from:

- code is required to be converted into machine code/binary
- code needs to be produced that can be understood by the computer

[1]

(c) Any **one** from:

- close to English/native/human language
- easier/faster to correct errors/read/write
- works on many different machines/operating systems (portable)

[1]

(e) 1 mark per correct letter, maximum 2 marks

Assembly code: **B**
High-level language code: **C**
Machine code: **A**

[2]

Q3)

Question	Answer	Marks
(a)	Any **two** from: • Close to the language processed by computers • May use mnemonics • An example is assembly language/machine code	**2**
(b)	Any **two** from: • Can directly manipulate the hardware • No requirement for the program to be portable • Program will be more **memory efficient** • No requirement for a compiler/interpreter • Quicker to execute • Can use specialised hardware	**2**

Q4)

Question	Answer	Marks
(a)	• C	**1**
(b)(i)	Any **three** from: • It translates the (high-level language) to low-level language/object code/machine code • It translates **all** the code **before** it is executed • It creates an executable file	**3**
(b)(ii)	Any **two** from: • It creates an error report after **trying to compile** • … displaying **all** errors in the code • … that require correction before execution can take place	**2**

Question	Answer	Marks
(c)	Any **three** from: e.g. • Code editors • Run-time environment • Built-in interpreter • Error diagnostics • Auto-completion • Auto-correction • Prettyprint	**3**

Q5)

Question	Answer	Marks
(a)	• B	**1**
(b)(i)	• Machine code // low-level language // object code	**1**
(b)(ii)	• Interpreter	**1**
(b)(iii)	• Compiler	**1**
(b)(iv)	• Compiler	**1**

Q6)

Question	Answer	Marks
6(a)	Any **one** from: • Computer system that is designed to a perform dedicated/single function • Computer system that contains a microprocessor (and)/dedicated hardware • Computer system that is built into a larger system	**1**

Question	Answer	Marks
6(b)	Assembler	**1**
6(c)	Any **one** from: • Convert to hexadecimal • Convert to denary • A character set can be used	**1**
6(d)	Any **one** from: • More control over manipulating the hardware • Faster execution for testing than a high-level language • They can use machine specific instructions	**1**
6(e)	**Six** from (MAX **4** for stating features): Code editor … … that allows the user to enter and amend code in their program Run-time environment … that allows a program to be run and see the outputs of their program Error-diagnostic … to show the programmer where there are errors in the program Auto-completion … to give the user **options/suggestions** of key commands to select Auto-correction … to correct a command that has a **minor misspelling** Prettyprint … changes the colour of **key commands** do they are easy to identify	**6**

Q7)

Question	Answer	Marks
7	Any **four** from: • It is a piece of software • … used to write/develop/edit code • … used test/debug the code • … with features such as auto-completion // any suitable example • To translate the code to low level language/machine code	**4**